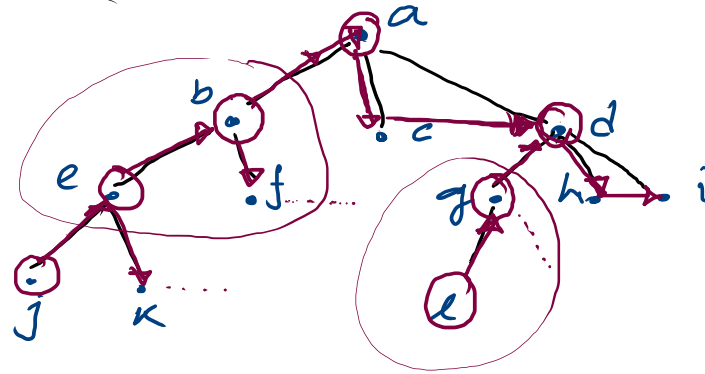


Chapter 11.2 Tree traversal

pre-order : **Root**, $L \rightarrow R$

in-order : **left most**, **Root**, $L \rightarrow R$

post order : $L \rightarrow R$, **Root**



Pre-order

→ a b c d
→ a b e f c d g h i
→ a b e j k f c d g h i

In-order

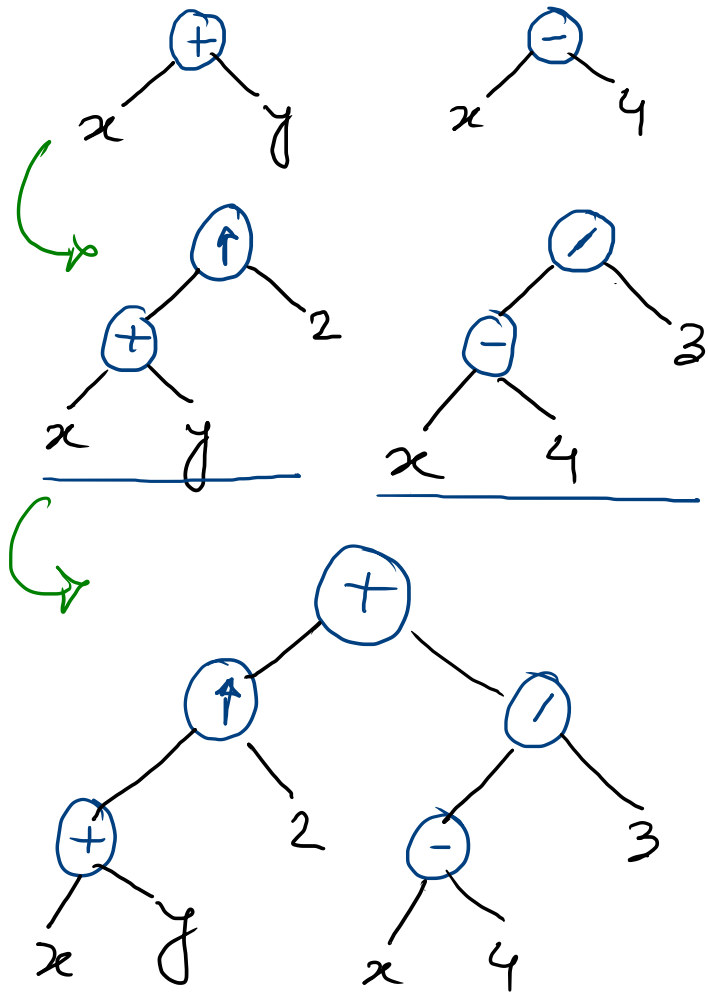
→ b a c d
→ e b f a c g d h i
→ j e k b f a c l g d h i

Post-order

→ b c d a
→ e f b c g h i d a
→ j k e f b c l g h i d a

⑧ Ordered rooted tree for expression

$$\rightarrow ((x+y)^2) + ((x-4)/3) =$$



- prefix (pre-order)

$+\uparrow/$
 $\rightarrow +\uparrow+2/-3$
 $\rightarrow +\uparrow+xy2/-x43$

- infix (in-order)

$\uparrow+$
 $\rightarrow +\uparrow2+/-3$
 $\rightarrow x+y\uparrow2+x-4/3$

- postfix (post-order)

$\uparrow/ +$
 $\rightarrow +2\uparrow-3/+$
 $\rightarrow xy+2\uparrow x4-3/+$

④ Evaluating postfix, prefix

④ Prefix: $R \rightarrow L$

$$+ - * 2 3 5 / \uparrow 2 3 4$$

$2 \uparrow 3 = 8$

$$+ - * 2 3 5 / 8 4$$

$8 / 4 = 2$

$$+ - * 2 3 5 2$$

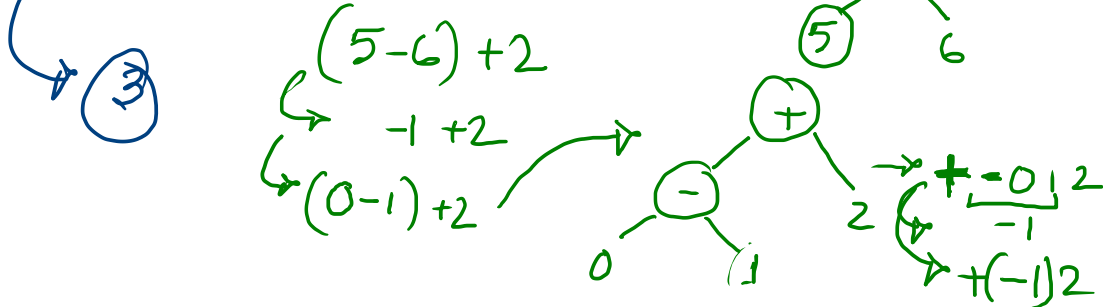
$2 * 3 = 6$

$$+ - 6 5 2 \rightarrow + - 5 6 2 \rightarrow (5 - 6) + 2 = -1 + 2 = 1$$

$6 - 5 = 1$ $5 - 6 = -1$

$$+ 1 2$$

$1 + 2 = 3$



④ Postfix: $L \rightarrow R$

$$7 \downarrow 2 \downarrow 3 * - 4 \uparrow 9 3 / +$$

$2 * 3 = 6$

$$7 6 - 4 \uparrow 9 3 / +$$

$7 - 6 = 1$

$$1 4 \uparrow 9 3 / +$$

$1^4 = 1$

$$1 9 3 / +$$

$9 / 3 = 3$

$$1 3 +$$

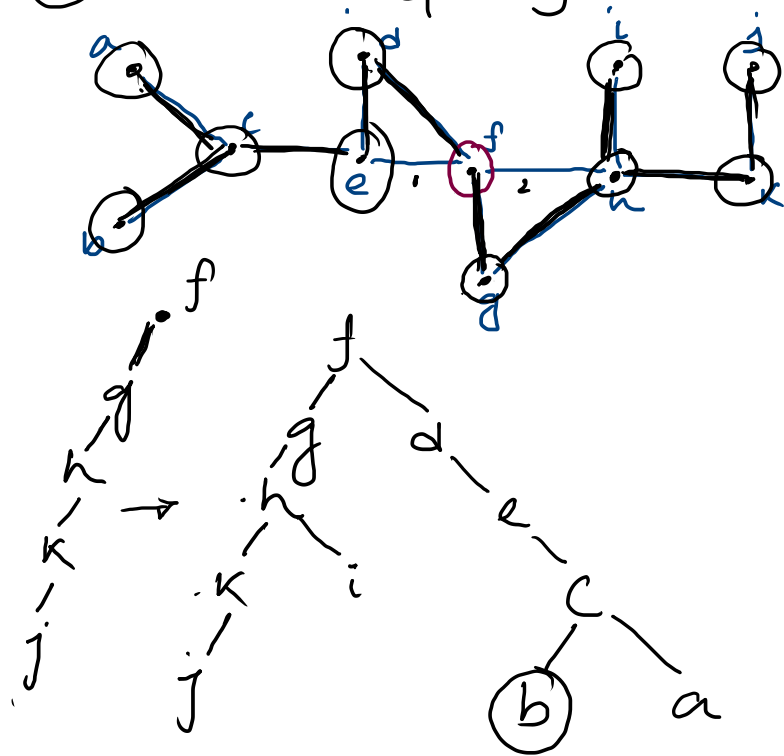
$1 + 3 = 4$

$$4$$

Chapter- 11.4 Spanning tree.

① **Spanning Tree**: the sub-graph of a graph G containing all the vertices of G .

② **DFS** $\rightarrow$ Depth first search



③ **BFS** - Breadth first search.

